ReSC-net: Hyperspectral Image Classification Based on Attention-Enhanced Residual Module and Spatial-Channel Attention

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***Abstract*— Hyperspectral image (HSI) classification is a key technique in remote sensing. Despite the increasing availability of high-quality HSI data, obtaining a large number of labeled samples remains challenging in certain cases. Consequently, HSI classification often faces the issue of insufficiently labeled samples. To address this challenge, sample augmentation techniques can be used to generate additional training samples. However, because of the significant differences between hyperspectral images and ordinary natural images, some augmentation techniques are not suitable for hyperspectral classification scenarios. In this article, considering phenomena such as spectral aliasing in hyperspectral image classification and imaging processes, we propose a novel online augmentation technique for hyperspectral samples. During training, we apply random gains to the center pixel of labeled samples to increase the number of usable samples. Additionally, since augmented samples may still be insufficient, using overly complex networks can lead to overfitting. Therefore, we introduce a hyperspectral image classification network called attention-enhancing residual and spatial-channel attention-based network (ReSC-net). In ReSC-net, we observe that the spatial dimension of hyperspectral blocks is much smaller than the channel dimension, and the limited sample size can lead to overfitting when using complex networks. Thus, we propose a channel attention-enhanced residual module to extract low-level features. Furthermore, ReSC-net introduces new spatial-channel attention to further optimize the extracted deep features for better classification. We conduct experiments on four commonly used HSI datasets. The experimental results demonstrate that our algorithm achieves favorable results on multiple HSI classification evaluation metrics.**

***Index Terms*— Attention-enhanced residual module, deep learning, hyperspectral image (HSI) classification, online sample amplification, spatial-channel attention.**

I. Introduction

HYPERSPECTRAL images contain a wealth of spectral information and have found extensive applications in remote sensing, including target detection [1], change detection [2], [3], urban planning [4], [5], and more. Among these applications, hyperspectral image classification, which involves assigning a unique label to each pixel, stands out as a critical technology [6]. However, the substantial spectral redundancy and ultrahigh data dimension inherent in hyperspectral image data present significant challenges in hyperspectral classification.

To address the challenges posed by the high dimensionality of hyperspectral data, a range of dimensionality reduction algorithms have been employed to explore the intrinsic structure of hyperspectral data in low-dimensional space. For instance, algorithms based on principal component analysis (PCA) [7] and linear discriminant analysis (LDA) [8], [9] were developed. These techniques aim to retain the intrinsic structure of hyperspectral image data or its low-dimensional embedding relationship in a reduced-dimensional space.

However, the performance of these approaches and models is somewhat limited due to their manual design, which can impact the accuracy of hyperspectral classification. Particularly in cases of limited labeled samples, the classification accuracy of these traditional algorithms is greatly diminished.

In recent years, with the rise of deep learning, numerous hyperspectral image classification algorithms based on deep learning have been proposed, achieving remarkable accuracy. These models include stacked autoencoders (SAEs) [10], [11], recurrent neural networks (RNNs) [12], [13], Bayesian approaches [14], graph convolutional networks (GCNs) [15], [16], [17], [18], [19], and transformers [20], [21], [22]. These algorithms mentioned above achieved higher classification accuracy compared to traditional hyperspectral algorithms. On this basis, some algorithms based on multiscale structure [23], 3-D-CNN [24], parallel dual-branch networks [25], [26], and attention-based networks [27], [28], [29], [30], [31].

The aforementioned deep learning-based algorithms have made significant advances in improving hyperspectral classification accuracy through adaptive end-to-end learning. However, even with these advancements, achieving high classification accuracy remains a challenge when dealing with limited samples. In practice, while hyperspectral image data are readily available, obtaining a sufficient number of representative and accurately labeled hyperspectral samples is still a daunting task. For instance, in remote areas such as uninhabited high-sea regions, acquiring a large number

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of labeled samples becomes nearly impossible for hyperspectral image resource exploration. Consequently, the issue of insufficient training sample size persists in hyperspectral classification.

The scarcity of labeled samples poses challenges, as complex networks may suffer from overfitting, resulting in subpar classification performance on the test dataset. Moreover, the high-dimensional nature of hyperspectral data and the unique characteristics of hyperspectral classification make it difficult to apply traditional data augmentation techniques used in natural images, such as cropping or linear addition, to hyperspectral classification tasks. As a result, hyperspectral image classification with a limited number of samples continues to face significant challenges. To address the challenge of achieving high accuracy with limited labeled training samples, researchers have proposed various solutions, including transfer learning algorithms [22], few-shot learning algorithms [32], [33], [34], [35], [36], self-supervised algorithms [37], [38], and cross-scene learning [39], [40]. In [32], researchers proposed a novel few-shot learning approach that uses geometric transformations, combined with a spectral-spatial feature extraction network and contrastive learning, to achieve better classification performance. In [34], researchers designed a novel spectral-spatial Siamese network, combined with weighted contrastive loss and adaptive cross-entropy, to enhance the discriminative power of the model, thus achieving effective classification under few-shot conditions. Furthermore, in RSEN [36], researchers proposed a robust self-ensembling network, which achieves self-ensembling by leveraging unsupervised learning from labeled and unlabeled data, allowing mutual learning between the base network and the ensemble network. This provides a new perspective on how to use unlabeled samples. However, this process is typically time-consuming. Additionally, in [41], researchers utilized preassigned classifiers combined with a simple classification network to achieve effective classification of hyperspectral images under limited labeled samples.

The algorithms mentioned above alleviate the problem of insufficiently labeled samples for hyperspectral image classification. However, most of these algorithms either require additional datasets for pretraining or rely on a large amount of unlabeled data for self-supervised learning, followed by fine-tuning on labeled samples [32], [36]. In addition, in practical applications, obtaining additional data or a substantial amount of unlabeled high-quality data is also a challenge.

To address the challenges mentioned above, we have proposed a novel online sample augmentation technique. This technique effectively increases the number of usable hyperspectral samples for learning without significantly increasing training time. Additionally, to efficiently learn features from limited samples in hyperspectral images and improve classification performance, we introduce a hyperspectral image classification network called residual and spatial-channel attention-based network (ReSC-net). Recognizing that overly complex networks struggle to capture the essence of hyperspectral image samples when data are scarce, the ReSC-net network utilizes channel attention-enhanced residual blocks to extract low-level features. Moreover, given that the patch size of hyperspectral inputs is much smaller than the number of channels, the network employs spatial-channel attention mechanisms to enhance feature representation. By combining sample enhancement techniques with an efficient feature extraction network, our proposed algorithm effectively mitigates the issue of limited sample size and achieves improved hyperspectral image classification results. The contributions of our research are summarized as follows.

1) To alleviate the issue of insufficient samples, we propose an online sample augmentation technique based on the central pixel. This technique involves mixing the spectra online to generate more training samples.
2) Building upon this, considering that the size of hyperspectral input patches is much smaller than the number of spectral bands, and complex networks tend to overfit when samples are scarce, we propose a residual module weighted by channel attention to extract low-level features.
3) Additionally, to enhance the classification performance of the network, we utilized a spatial-channel attention mechanism to optimize the low-level features extracted by the previous module.
4) Based on the aforementioned modules, we propose a new hyperspectral classification network called ReSC-net. We evaluated the performance of our classification algorithm on four widely used hyperspectral image classification datasets. The experimental results demonstrate that our algorithm surpasses state-of-the-art methods, achieving superior classification performance across all four datasets.

The remainder of this article is organized as follows. Section II provide0s a comprehensive explanation of the proposed algorithms and details their key components and methodologies. In Section III, we showcase the experimental results derived from conducting experiments on four widely utilized hyperspectral datasets. The performance of our algorithms is evaluated using various metrics, and comparisons with existing approaches are drawn. At last, Section IV delves into the conclusions drawn from our study, summarizes the contributions made, and offers insights into potential future research directions.

II. Proposed Method

The proposed ReSC-net framework is illustrated in Fig. 1. As shown in the figure, the entire HSI classification framework consists mainly of three main modules: online sample amplification, attention-enhanced residual block, and spatial-channel attention module. In the following subsection, we will analyze each of these modules.

A. Online Central Pixel-Based Sample Amplification

Although the accessibility of hyperspectral remote sensing image data has improved significantly, challenges remain to acquire a large number of labeled hyperspectral remote sensing images. For example, in mineral exploration conducted using hyperspectral images in remote high-altitude areas, it is

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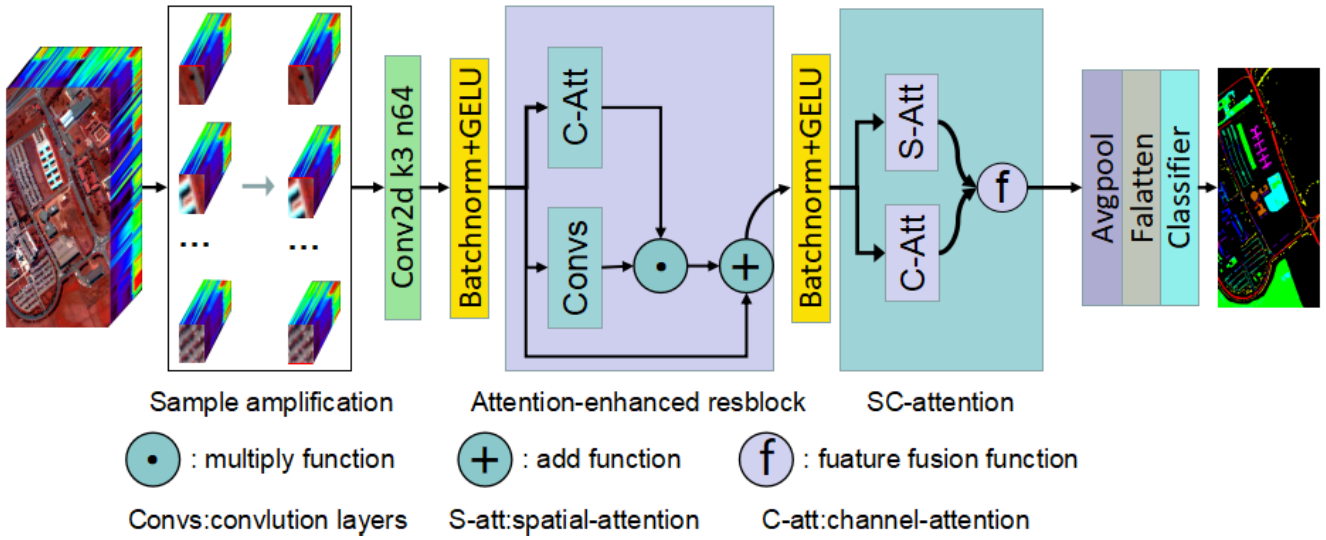


Fig. 1. Proposed ReSC-net. The ReSC net contains three main parts, a sample amplification module, an attention-enhanced resblock, and an SC-attention.

often difficult to collect a sufficient number of hyperspectral image samples. In situations with limited samples, sample-augmentation techniques can enhance the performance of neural networks, such as through methods like crop rotation. However, traditional data augmentation methods for natural images may not be directly applicable to hyperspectral image classification scenarios due to the unique properties of hyperspectral images. One common phenomenon in hyperspectral imaging is pixel aliasing, which arises from insufficient spatial resolution. An aliased pixel refers to a single pixel that is a mixture of multiple solitary pure pixels with different proportions of noise. During classification, labels are typically assigned based on the dominant features. For example, if a pixel is predominantly composed of features of Class A but contains 10% of the signal from Class B, it would still be labeled as Class A. Aliased pixels and noise contribute to isotropic phenomena commonly observed in hyperspectral images.

Given the significant impact of spectral signals on the accuracy of hyperspectral image classification, we propose a novel approach that utilizes spectral aliasing to generate a new set of spectral pixels. This technique aims to increase the training sample size and improve the accuracy of hyperspectral image classification. Our sample augmentation technique is tailored for classification algorithms that utilize block-based extraction of spatial and spectral information from hyperspectral images. The workflow of our proposed sample amplification method is illustrated in Fig. 2.

As illustrated in Fig. 2, let us consider a subhyperspectral image patch **HSI**, which is a tensor cube with a size of $[C,\ P,\ P]$, where C represents the number of channels and P is the patch size used during training. The spatial location of the center pixel in this patch is denoted as $\mathbf{HSI}(:, (p/2), (p/2))$. In hyperspectral imaging, because of the presence of mixed pixel cells, we can consider constructing new samples by utilizing the pixels surrounding the intermediate pixels to represent the aliasing effect. Mathematically,

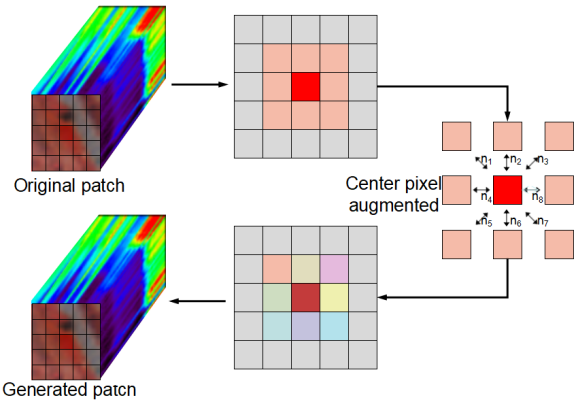


Fig. 2. Sample amplification model. The center pixel with the training sample block will regain the value of its surrounding pixel block multiplied by a smaller random number and then linearly mixed.

this can be expressed as follows:

$$\mathbf{HSI}\left(:, \frac{p}{2}, \frac{p}{2}\right) = \sum_{i=-1}^{1} \sum_{j=-1}^{1} n_{i,j} \mathbf{HSI}\left(:, \frac{p}{2}+i, \frac{p}{2}+j\right). \quad (1)$$

In this way, the center pixel data contains blending information from the eight surrounding pixels. However, to ensure that the label of the hyperspectral patch remains consistent with the original label, it is important to preserve as much of the original spectral information as possible. Therefore, the weights assigned to the surrounding pixels should be appropriately reduced.

Furthermore, it is worth noting that the original pixels may already exhibit some degree of spectral aliasing during the imaging process. To address this issue, we can subtract the spectral value of the original pixel from the spectral values of the surrounding pixels. This purification process helps alleviate the spectral aliasing effect and enables some degree of sample amplification. As a result, the weight $n_{i,j}$ in 1 should be a small positive or negative value. To maintain the original spectral

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variation while ensuring it remains within a reasonable range, the weight $n_{i,j}$ should satisfy the following conditions:

$$\begin{cases} \sum_{i,j} n_{i,j} == 1 \\ n_{1,1} > \sum n_{i,j}, \quad [i,j] \neq [1,1]. \end{cases} \tag{2}$$

In this article, we utilize randomly generated weights ranging from -0.3 to 0.3 for the surrounding pixels, excluding the center pixel. However, if the absolute value of the sum of these weights exceeds 0.4, we apply a scaling factor of 0.4 to ensure that the overall weight remains within this range. This approach helps control the magnitude of the weights and maintains balance in the amplification process.

B. Attention-Enhanced Residual Model

ResNet [42] has made significant advances in various computer vision tasks and has been widely adopted in remote sensing image processing. The effectiveness of simple residual connections in neural networks, especially in deep architectures, has been well established. However, in hyperspectral image classification, the limited number of available samples can lead to overfitting issues when using deep neural networks.

To address this challenge, our proposed network employs simplified residual modules. In hyperspectral image input blocks, the channel dimension of input features is typically much larger than the spatial dimension. Therefore, to enhance the feature extraction capability of individual residual modules, we introduce a channel attention module to enhance the nonlinear representation power of the residual module. This modified residual module, referred to as the attention-enhanced residual block, improves the feature representation and discrimination capability in hyperspectral image classification, thereby enhancing classification performance.

In this article, we incorporate the channel attention mechanism, also known as efficient channel attention (ECA), proposed in [43], to enhance the Resblock in our network. This modification can be visualized in Fig. 3.

As illustrated in the figure, we extract the channel attention information, referred to as "**att**," from the input of the Residual Block. Mathematically, this can be expressed as follows:

$$\mathbf{att} = \delta\left(\mathcal{C}_{1\mathrm{D},k}(\mathcal{A}(\mathbf{X}))\right) \tag{3}$$

where $\mathcal{A}$ means adaptive average pooling 2-D, $\mathcal{C}_{1\mathrm{D},k}$ means the 1-D convolution with a kernel size of k, and δ is the sigmoid function. The abstracted attention information will be multiplied by the abstracted residual information and then added to the original input, which can be expressed as follows:

$$\hat{\mathbf{X}} = \mathcal{F}(\mathbf{X}) \times \mathbf{att} + \mathbf{X} \tag{4}$$

where the $\mathcal{F}$ consists of three 2-D convolution layers with 2-D batch norm and ReLu function. $\hat{\mathbf{X}}$ is the output of this module.

C. Spatial-Channel Attention Model

Although the residual block can extract enhanced features for classification tasks, we opt for a relatively straightforward

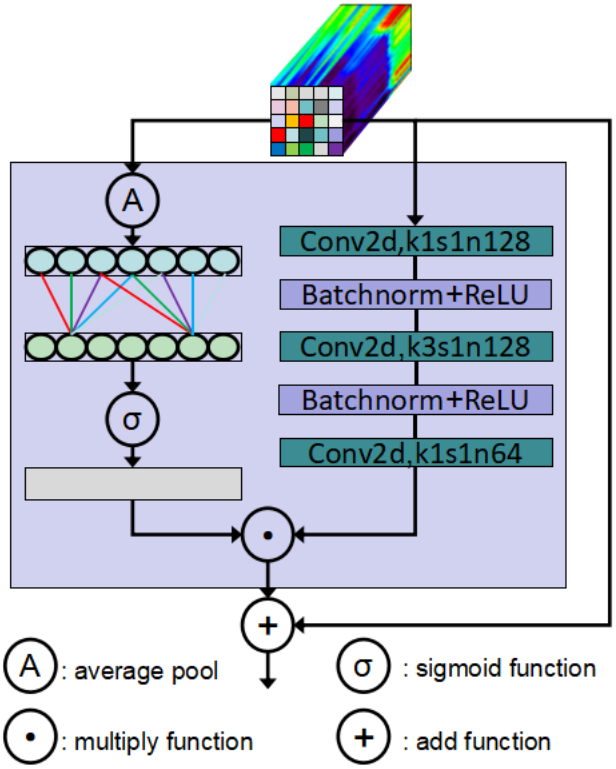


Fig. 3. Attention-enhanced residual module. In this module, the attention weight values of each channel are extracted from the input features, which are weighted by dot-multiplication with the convolved features, and then a residual join is used to sum the input values with the weighted values to get the final output.

variant to alleviate the risk of overfitting. Consequently, the features extracted by this module can be regarded as low-level features. To further boost the classification performance, we introduce a fusion of spatial-channel attention modules for hyperspectral image classification.

The spatial-channel attention modules are crafted to capture and accentuate the crucial spatial and channel-wise information within the extracted features. By amalgamating spatial and channel attention mechanisms, our proposed approach aims to enhance the feature's expressive capability and augment their discriminative power for classification tasks.

The spatial-channel attention model is depicted in Fig. 4. The channel attention mechanism that we utilized adopts the ECA attention, similar to the equation presented in 3. When obtaining channel attention weights **att** from the input of this module, they can be element-wise multiplied to each channel's feature map to obtain the feature representation weighted by channel attention.

For spatial attention, we exploit the spatial relationships among features. Unlike channel attention, spatial attention emphasizes spatial location information, serving as a complement to channel attention. Our spatial attention module resembles that in [44]. To compute spatial attention, we initially employ average pooling and maximum pooling operations along the channel axis, followed by concatenation to generate an initial feature representation. This representation undergoes convolution via a standard convolutional layer,

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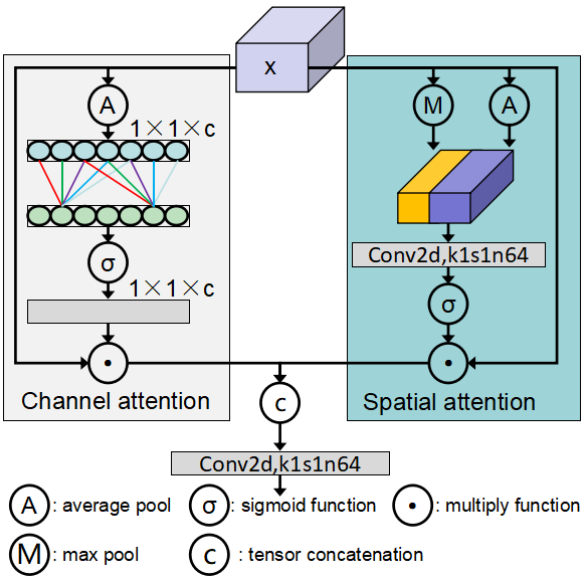


Fig. 4. Spatial-channel attention model. In this module, the input features are attentively weighted in spatial and channel dimensions, respectively, and then concatenated, and then fed into a 2-D convolutional layer that has acquired the null-spectrum weighted feature values. c means the channel number of the input feature.

succeeded by activation through the sigmoid function to yield the 2-D spatial attention map. The computation of spatial attention can be expressed as follows:

$$\mathbf{X_{max}} = \mathcal{M}\mathrm{ax}\left(\hat{\mathbf{X}}\right) \tag{5}$$

$$\mathbf{X_{mean}} = \mathcal{M}\mathrm{ean}\left(\hat{\mathbf{X}}\right) \tag{6}$$

$$\mathbf{X_{att}} = \delta\left(\mathcal{C}_{2\mathrm{D},k}([\mathbf{X_{max}}, \mathbf{X_{mean}}])\right) \tag{7}$$

where $\hat{\mathbf{X}}$ represents the feature map obtained from the previous model. The operations of $\mathcal{M}$ax and $\mathcal{M}$ean refer to the maximum and average pooling operations along the dimension 1, respectively. [,] denotes the concatenation operation along dimension 1. $\mathcal{C}_{2\mathrm{D},k}$ represents a 2-D convolutional layer with a kernel size of k and δ represents the sigmoid function.

We then perform element-wise multiplication of the extracted channel attention and spatial attention with the input features, concatenate them into a longer tensor along dimension 1, and restore the tensor to the size of the input using a 2-D convolutional layer with a sigmoid function. This process can be expressed as follows:

$$\mathbf{C_{att}} = \hat{\mathbf{X}} \times \mathbf{att} \tag{8}$$

$$\mathbf{S_{att}} = \hat{\mathbf{X}} \times \mathbf{X_{att}} \tag{9}$$

$$\mathbf{Att_f} = \mathcal{C}\mathrm{at}(\mathbf{C_{att}}, \mathbf{S_{att}}) \tag{10}$$

$$\mathbf{Y} = \delta\left(\mathcal{C}_{2\mathrm{D},k}(\mathbf{Att_f})\right) \tag{11}$$

where **att** represents the channel attention feature obtained from $\hat{\mathbf{X}}$, and $\mathbf{Y}$ is the output feature. $\mathbf{C_{att}}$, $\mathbf{S_{att}}$, and $\mathbf{Att_f}$ denote channel attention, spatial attention, and fused attention, respectively. The operation $\mathcal{C}$at represents concatenation along the channel dimension.

D. Loss Function

Based on the above, we construct ReSC-net for hyperspectral classification. The network incorporates the online sample enhancement module during training, which augments the training data. However, this module is not utilized during testing.

In this article, we employ the cross-entropy loss directly as the final loss function. The cross-entropy loss function quantifies the disparity between the predicted probability distribution and the true labels of the samples. Assuming we have N samples and T classes, and each sample's true label is encoded using one-hot encoding, the cross-entropy loss can be mathematically expressed as follows:

$$\mathbf{Loss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{T}\mathbf{y}_{ij}\log\left(\hat{\mathbf{y}}_{ij}\right) \tag{12}$$

where $\mathbf{y}_{ij}$ represents the jth element of the true label for sample i, and $\hat{\mathbf{y}}_{ij}$ represents the predicted probability of sample i belonging to class j. The summations are performed overall N samples and T classes. The cross-entropy loss is calculated by taking the negative of the average of the cross-entropy between the predicted probability distribution and the true labels for all samples.

III. Experiment and Analysis

To evaluate the effectiveness of the proposed algorithm, we conducted experiments and analysis on several HSI datasets, namely Indian pines (IPs), Pavia University (PU), Kennedy Space Center (KSC), and WHU-Hi-HongHu (Honghu/HH) [45]. IP, PU, and KSC can be found on the website.[1] The Honghu dataset can be found on the website.[2] The landcover detail and experimental settings for the four datasets can be found in Tables I–IV.

A. Dataset Description

1) Indian Pines: The IP dataset was acquired using the airborne visible/infrared imaging spectrometer (AVIRIS) sensor on the Indiana Pines agricultural test site in June 1992. The original dataset consists of 220 bands, but researchers removed some noise-affected and water-absorption bands. The processed dataset used in our experiment contains 200 bands with a size of 145 × 145, covering a wavelength range of 0.4–2.5 μm. There are 16 vegetation classes included in this dataset, with a total of 10 249 labeled pixels. The spatial resolution of this dataset is 20 m.

2) Pavia University: The PU dataset was collected over PU in Northern Italy using the reflective optics system imaging spectrometer (ROSIS) during a flight campaign in 2001. After removing 12 noisy bands, the dataset consists of 103 bands with a size of 610 × 340. The spatial resolution of this dataset is 1.3 m per pixel, and the wavelength range is 0.43–0.86 μm. The dataset contains a total of 42 776 labeled pixels belonging to nine urban land cover categories.

3) Kennedy Space Center: The KSC dataset was acquired using the AVIRIS instrument over the KSC in the USA in 1996. After removing water absorption and noisy bands, the dataset consists of 176 bands for analysis. The spatial resolution of this dataset is 18 meters per pixel, and it has a

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[2] http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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TABLE I
CATEGORY INFORMATION OF THE IPS DATASET

Class	Color	Class Name	Training	Validation	Test
1		Alfalfa	10	5	43
2		Corn-notill	10	5	1369
3		Corn-mintill	10	5	795
4		Corn	10	5	226
5		Grass-pasture	10	5	462
6		Grass-trees	10	5	699
7		Grass-pasture-mowed	10	5	25
8		Hay-windrowed	10	5	457
9		Oats	10	5	17
10		Soybean-notill	10	5	951
11		Soybean-mintill	10	5	2354
12		Soybean-clean	10	5	568
13		Wheat	10	5	194
14		Woods	10	5	1212
15		Building-Grass-Trees-Drives	10	5	369
16		Stone-Steel-Towers	10	5	88
		Total	160	80	10009

TABLE II
CATEGORY INFORMATION OF THE PU DATASET

Class	Color	Class Name	Training	Validation	Test
1		Asphalt	10	5	6616
2		Meadows	10	5	18634
3		Gravel	10	5	2084
4		Trees	10	5	3049
5		Painted-m-s	10	5	1330
6		Bare-soil	10	5	5014
7		Bitumen	10	5	1315
8		Self-block-b	10	5	3667
9		Shadows	10	5	932
		Total	90	45	42641

TABLE III
CATEGORY INFORMATION OF THE KSC DATASET

Class	Color	Class Name	Training	Validation	Test
1		Scrub	10	5	44
2		Willow-swamp	10	5	1370
3		Cp-hammock	10	5	796
4		Cp/O-hammock	10	5	227
5		Slash-pine	10	5	463
6		Oak/B-Bammock	10	5	700
7		HW-swanp	10	5	26
8		Graminoid-marsh	10	5	458
9		Spartine-marsh	10	5	18
10		Cattail-Marsh	10	5	952
11		Salt-marsh	10	5	2355
12		Mud-flats	10	5	569
13		Water	10	5	195
		Total	130	75	5006

size of 512 × 614 pixels. The wavelength range of the dataset is from 0.4 to 2.5 μm. There are a total of 5211 labeled pixels in the dataset, representing 13 classes of upland and wetland areas.

4) WHU-Hi-HongHu: The Honghu/HH dataset was collected using a 17-mm focal length Headwall Nano-Hyperspec

TABLE IV
CATEGORY INFORMATION OF THE HONGHU/HH DATASET

Class	Color	Class Name	Training	Validation	Test
1		Red roof	10	5	14036
2		Road	10	5	3497
3		Bare soil	10	5	21806
4		Cotton	10	5	163270
5		Cotton firewood	10	5	6203
6		Rape	10	5	44542
7		Chinese cabbage	10	5	24088
8		Pakchoi	10	5	4039
9		Cabbage	10	5	10804
10		Tuber mustard	10	5	1279
11		Brassica parachinensis	10	5	11000
12		Brassica chinensis	10	5	8939
13		Small Brassica chinensis	10	5	22492
14		Lactuca sativa	10	5	7341
15		Celtuce	10	5	987
16		Film covered lettuce	10	5	7247
17		Romaine lettuce	10	5	2995
18		Carrot	10	5	3202
19		White radish	10	5	8697
20		Garlic sprout	10	5	3471
21		Broad bean	10	5	1313
22		Tree	10	5	4025
		Total	220	110	384119

imaging sensor mounted on a DJI Matrice 600 Pro UAV platform on November 20, 2017, in Honghu City, Hubei province, China. This dataset is a high-resolution hyperspectral remote sensing image dataset captured by unmanned aerial remote sensing. It has a spatial resolution of approximately 0.043 m and a size of 940 × 475 pixels. The dataset consists of 270 bands, covering a spectral range from 0.4 to 1 μm. There are 22 different feature categories represented in this dataset.

B. Evaluation Metrics

To compare the classification performance of different algorithms and analyze the impact of various parameters, we conducted both quantitative and subjective evaluations. For quantitative assessment, we utilized various metrics, including overall accuracy (OA), average accuracy (AA), kappa coefficient (Kappa), and accuracy of each class (AEC). OA indicates the ratio of correctly classified samples to the total number of samples, while AA represents the AA across all land cover categories. The Kappa coefficient offers an overall measure of agreement between the classification results and the actual categories. AEC measures the accuracy of each individual category, which is particularly valuable for unbalanced datasets.

In addition to quantitative evaluation, we performed subjective visual assessments by visualizing the classification results of each algorithm and comparing them with the ground truth labels. This enabled a direct visual comparison of the performance and effectiveness of different algorithms.

To ensure a fair comparison, we also considered additional factors such as training time, testing time, number of model parameters, and computational efficiency measured in flops.

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TABLE V
QUANTITATIVE CLASSIFICATION RESULTS OF THE IPS DATASET WITH TEN LABELED TRAINING SAMPLES PER CLASS. BEST RESULTS ARE SHOWN IN BOLD

Class	RF	SVM	ContextNet	SSAN	RSSAN	SSAtt	A2S2K	SSTN	SSSAN	CVSSN	RSEN	ETF	Proposed
1	29.01±5.82	18.56±3.53	38.29±22.18	45.45±23.04	31.63±9.93	80.67±10.20	47.59±20.19	73.12±15.22	65.67±11.99	71.17±16.24	**98.61±1.86**	73.02±12.19	65.28±19.44
2	42.67±9.52	43.55±2.73	45.51±13.87	45.89±10.56	44.07±8.27	66.84±6.35	72.06±7.54	78.23±8.80	58.88±11.23	73.18±7.15	37.78±12.33	82.72±4.84	**84.28±10.39**
3	37.56±13.11	33.71±8.46	38.77±11.45	45.99±13.41	33.21±4.24	67.76±10.27	60.54±14.82	66.36±11.55	57.60±6.76	71.86±11.27	50.46±16.66	81.15±11.67	**83.17±8.61**
4	19.65±4.77	24.60±3.10	26.77±18.70	45.59±17.38	20.86±7.19	56.80±9.05	57.40±13.21	67.77±9.75	58.86±8.82	64.40±9.88	67.67±18.45	**73.98±10.26**	73.97±9.42
5	63.78±8.17	62.30±7.74	63.08±14.92	54.48±11.41	51.03±7.69	83.28±15.09	92.27±4.04	93.42±3.03	85.95±16.70	92.06±4.52	73.70±15.68	**95.08±4.60**	94.82±3.38
6	82.36±4.14	87.20±3.73	78.28±19.81	73.44±6.18	67.20±7.66	94.26±4.90	95.82±2.87	95.16±2.61	93.54±4.06	96.00±1.59	**97.10±3.72**	93.87±3.03	95.17±2.67
7	13.49±9.05	22.76±11.87	19.26±13.55	39.59±17.70	18.38±4.88	21.45±11.71	40.11±27.52	23.41±9.01	30.33±16.96	37.56±12.67	**99.44±1.67**	41.43±18.30	31.08±12.90
8	90.78±5.47	96.48±1.29	92.58±15.11	96.76±3.23	90.47±5.75	99.06±0.74	93.48±8.72	97.52±1.72	98.73±1.28	98.26±2.02	**100.00±0.00**	99.04±1.67	98.42±2.30
9	2.89±1.03	8.41±3.90	8.22±8.66	11.81±7.53	4.83±1.52	9.43±7.50	10.58±5.50	8.77±9.19	15.59±10.91	19.83±13.91	**100.00±0.00**	19.42±14.05	10.92±5.56
10	42.98±0.84	49.44±6.21	44.32±13.44	48.51±9.70	38.68±3.84	57.52±8.80	56.92±10.62	66.50±8.63	51.24±4.03	66.58±10.40	69.84±8.47	76.74±10.58	**81.12±12.55**
11	63.12±4.90	65.89±4.84	65.05±3.88	73.84±10.11	65.48±4.82	77.64±6.40	78.16±8.29	89.89±5.91	74.33±6.80	83.25±6.79	79.64±15.70	**92.23±3.74**	91.52±6.82
12	26.75±6.43	32.32±3.28	34.65±7.97	36.87±7.61	31.06±6.85	48.51±7.16	52.46±14.19	66.46±12.86	44.75±12.58	65.06±5.75	45.64±16.05	79.38±12.28	**84.70±13.16**
13	76.68±4.31	83.43±3.46	59.04±19.36	66.62±19.50	57.25±10.11	84.23±9.35	84.11±7.22	85.11±7.31	95.11±6.04	88.85±10.23	**99.49±0.40**	83.36±7.59	80.19±11.74
14	91.09±3.81	90.20±4.56	80.57±9.16	88.43±3.52	85.28±3.19	95.56±2.28	94.80±4.60	97.27±1.15	93.69±6.48	97.69±1.32	90.57±9.62	95.89±1.95	**97.91±1.62**
15	38.34±6.67	42.41±7.98	51.98±16.08	69.09±9.89	46.82±12.89	70.02±7.69	69.00±15.92	76.09±13.56	70.25±9.89	74.50±8.89	**81.57±8.44**	77.71±10.70	78.82±12.97
16	64.70±18.64	87.63±10.64	69.09±13.15	56.71±11.64	52.67±23.78	70.92±8.75	48.40±25.97	65.59±11.78	79.15±6.29	70.03±12.16	**97.35±3.95**	61.77±12.43	67.88±8.98
OA	52.54±2.62	55.42±1.86	51.27±6.92	57.87±4.33	52.19±1.99	72.78±2.70	70.59±7.59	79.66±1.52	69.62±2.88	79.11±0.86	72.21±3.75	85.40±1.57	**85.98±1.93**
AA	49.12±2.10	53.06±1.35	50.97±4.70	56.19±5.30	46.18±3.19	67.75±1.78	65.86±5.07	71.92±2.45	67.10±2.14	73.14±1.60	**80.55±1.81**	76.68±2.68	76.20±2.38
Kappa	46.81±2.74	50.07±2.01	45.74±6.94	52.88±4.79	46.40±2.01	69.28±2.87	66.93±8.46	77.09±1.69	65.64±3.32	76.39±0.98	68.15±3.94	83.46±1.75	**84.08±2.13**

C. Experimental Setting

The experiments were conducted on a workstation equipped with an Intel[3] Xeon[3] Silver 4210 CPU running at 2.20 GHz and a single NVIDIA GeForce RTX 2080 GPU with 11 GB of GPU memory. The system environment used was Ubuntu [insert corrected version number here]. The proposed algorithm was implemented in Python 3.6 using the PyTorch 1.10 framework. During training, Adam Optimizer was utilized with a fixed learning rate of 0.01, and a batch size of 32 was used for training on all datasets.

Since our algorithm incorporates random data augmentation during training, we conducted each experiment for 150 epochs to ensure convergence. To ensure a fair comparison, all algorithms were repeated randomly ten times and the average result and variance were calculated as the final measurement result. In the table results, the best results are highlighted in bold, while the second-best results are underlined for clear comparisons.

For patch-based HSIC algorithms, the spatial window size of the 3-D patch was fixed at 11×11. This size was chosen based on previous research and considerations regarding computational efficiency and performance. It is important to note that all algorithms utilized the same training and test datasets to ensure consistency in the experimental setup.

D. Comparative Experiments

In order to evaluate the effectiveness of the proposed algorithm, we conducted experiments comparing it with ten benchmark methods. These methods include two classic machine learning-based algorithms and eight state-of-the-art deep learning algorithms. The classic machine learning methods selected for the experiment include random forest (RF) [46] and support vector machine (SVM) with a radial basis function (RBF) [47]. The deep learning-based models include ContextNet [48],[4] SSAN [28], RSSAN [30],[5] SSAtt [49],[6] A2S2K [27],[7] SSTN [29],[8] SSSAN [26], CVSSN [50],[9] RSEN [36],[10] and ETF [41].

The selected comparison algorithms are based on the study by [50], which provides a comprehensive coverage of popular approaches to the classification of hyperspectral images. These include classic machine learning methods, 2-D CNN-based algorithms, 3-D CNN-based algorithms, residual learning methods, and various attention-based algorithms. It should be noted that RSEN and ETF are specifically designed for hyperspectral image classification with limited labeled samples. To ensure fair comparisons, all methods were trained using the same settings.

1) Quantitative Accuracy Analysis: The quantitative results are reported in Tables V–VIII. In these tables, we compare our algorithm with others in terms of classification performance on four datasets using metrics such as average classification AEC, OA, AA, and Kappa coefficients.

Observing these tables, it is evident that certain nondeep learning algorithms outperform specific deep learning-based algorithms on particular datasets. For example, on the IPs dataset, RF and SVM exhibit significantly better classification performance than ContextNet and RSSAN. This outcome can be attributed to the scarcity of training samples, making it challenging for some deep learning algorithms to learn robust classification features or avoid overfitting on the limited training set, resulting in suboptimal classification results. In contrast, traditional machine learning algorithms demonstrate relatively better performance in handling small sample sizes, thus achieving superior classification performance. A similar trend is observed in the KSC dataset.

[3]Registered trademark.

[4]https://github.com/eecn/Hyperspectral-Classification
[5]https://github.com/liererrniu/RSSAN-Hyperspectral-Image
[6]https://github.com/weecology/DeepTreeAttention
[7]https://github.com/suvojit-0X55aa/A2S2K-ResNet
[8]https://github.com/zilongzhong/SSTN
[9]https://github.com/lms-07/CVSSN
[10]https://github.com/YonghaoXu/RSEN

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TABLE VI

QUANTITATIVE CLASSIFICATION RESULTS OF THE PU DATASET WITH TEN LABELED TRAINING SAMPLES PER CLASS. BEST RESULTS ARE SHOWN IN BOLD

Class	RF	SVM	ContextNet	SSAN	RSSAN	SSAtt	A2S2K	SSTN	SSSAN	CVSSN	RSEN	ETF	Proposed
1	87.90±6.24	92.14±7.28	88.51±6.46	79.14±1.68	83.56±6.12	93.32±3.63	95.30±3.18	96.00±2.93	95.93±3.46	96.64±3.81	84.69±15.97	**96.86±2.11**	96.70±2.73
2	84.58±2.45	86.72±3.74	91.30±2.26	92.03±1.86	88.45±3.08	94.38±2.41	96.59±2.45	96.95±1.06	91.47±3.70	**99.95±2.11**	85.75±15.04	98.28±1.78	99.23±0.90
3	35.62±4.16	37.22±3.77	39.60±8.07	36.17±8.78	37.67±6.25	62.45±12.45	60.49±16.04	87.33±8.38	62.35±11.71	78.90±5.78	86.08± 11.40	**88.02±9.27**	85.52±9.95
4	56.57±6.76	57.60±9.59	76.74±20.73	79.24±1.17	79.96±16.15	85.40±14.70	85.31±14.10	86.67±18.25	82.33±23.19	91.21±14.27	**98.82± 1.01**	87.08±15.39	85.95±17.10
5	92.06±2.01	95.07±2.43	86.37±12.86	94.35±1.17	85.19±10.88	97.09±4.01	96.39±3.60	98.15±2.54	96.80±1.87	**98.65±2.77**	98.07±2.57	98.14±3.33	96.31±3.66
6	33.75±4.89	6.66±7.99	41.05±5.81	48.07±1.17	43.59±8.28	56.16±9.00	69.00±10.02	68.53±16.86	54.54±9.29	68.54±9.89	58.56±15.31	75.20±13.64	**78.80±12.55**
7	43.94±4.20	38.03±4.47	35.75±10.42	40.30±1.17	34.37±10.03	53.18±17.27	56.03±18.12	83.98±12.52	59.83±10.71	65.90±15.28	65.83 ±16.94	78.62±10.55	**92.18±7.77**
8	65.90±8.17	66.33±5.61	71.79±8.87	64.65±1.17	62.37±8.44	75.87±4.46	75.09±10.52	84.73±6.25	80.30±4.35	84.01±3.76	72.07± 17.77	87.90±3.58	**89.52±4.44**
9	98.72±1.00	**99.66±0.23**	54.15±25.54	76.74±1.17	83.63±11.53	89.63±8.45	96.37±3.65	93.48±3.35	98.56±1.85	97.51±3.50	95.44± 7.02	93.41± 2.83	93.36±8.76
OA	63.66±3.77	62.77±6.36	65.16±6.38	68.14±12.20	67.68±4.92	80.20±5.44	83.04±5.16	87.28±6.20	79.24±3.38	87.70±3.60	82.15± 6.94	90.27±4.71	**91.29±4.74**
AA	66.56±2.02	67.71±1.04	65.03±3.75	67.85±6.52	66.53±4.62	78.61±4.78	81.17±4.51	88.42±3.48	79.88±3.35	86.48±3.18	82.81± 3.05	89.28±3.53	**90.73±3.74**
Kappa	55.17±3.78	54.67±5.70	57.15±6.45	60.32±13.45	58.95±5.36	74.74±6.50	78.28±6.24	83.85±7.33	73.40±3.94	84.10±4.54	76.92± 8.20	87.49±5.76	**88.86±5.74**

TABLE VII

QUANTITATIVE CLASSIFICATION RESULTS OF THE KSC DATASET WITH TEN LABELED TRAINING SAMPLES PER CLASS. BEST RESULTS ARE SHOWN IN BOLD

Class	RF	SVM	ContextNet	SSAN	RSSAN	SSAtt	A2S2K	SSTN	SSSAN	CVSSN	RSEN	ETF	Proposed
1	93.73±3.81	74.52±2.51	96.49±4.03	95.45±1.68	94.85±3.78	98.14±1.99	97.32±2.20	98.48±3.51	96.31±3.60	99.38±0.83	81.64±7.43	99.36±0.63	**99.80±0.40**
2	69.69±4.48	84.26±6.39	65.06±8.02	74.75±1.86	62.30±22.20	88.54±7.55	76.35±22.64	95.67±4.03	79.86±9.88	95.50±6.03	63.13±9.65	**98.29±1.97**	97.85±2.82
3	83.14±2.61	66.74±6.93	69.04±7.05	59.65±8.78	57.12±8.32	77.46±6.83	74.74±31.16	86.22±20.09	64.47±13.41	90.87±8.35	85.00±8.07	90.30±10.66	**97.82±4.39**
4	55.81±6.19	44.72±5.93	60.87±8.21	58.01±3.24	49.59±14.71	72.97±10.93	64.72±20.14	75.04±27.41	58.74±12.55	84.65±8.16	46.94±11.65	**86.10±15.11**	78.67±29.82
5	54.71±8.69	47.42±7.71	8.99±7.96	47.20±1.68	45.15±8.89	67.82±10.78	69.36±25.52	67.92±20.51	45.26±15.72	69.50±15.70	**74.24±5.40**	73.65±18.77	71.47±21.75
6	43.88±5.74	45.87±8.98	80.57±7.34	56.94±1.86	58.45±14.00	82.13±14.42	91.97±8.50	68.03±28.95	69.69±14.78	86.38±11.85	48.72±9.60	82.84±14.59	**94.54±8.66**
7	60.33±6.7	68.35±4.44	68.17±16.52	53.12±8.78	47.61±14.54	71.41±12.78	77.15±20.26	69.85±18.07	62.99±11.25	70.62±13.50	**88.11±8.83**	69.34±20.70	84.40±14.72
8	51.90±5.73	61.23±5.50	66.54±11.0	65.07±3.24	52.80±16.76	90.12±4.14	82.62±11.70	84.04±7.99	58.26±14.51	77.37±19.22	68.65±13.75	91.05±10.12	**97.68±4.89**
9	75.51±5.05	75.00±3.73	86.68±7.76	74.45±1.68	65.78±9.84	87.51±9.22	91.16±11.07	96.51±4.73	74.47±8.83	90.22±8.99	79.27±11.73	94.69±8.06	**99.09±2.28**
10	73.10±9.74	90.05±9.01	67.55±11.56	2.99±1.86	69.85±10.22	94.67±5.33	92.45±12.05	98.12±3.41	67.76±11.25	85.41±30.26	57.36±12.42	**100.00±0.00**	**100.00±0.00**
11	96.39±3.74	92.42±2.97	95.02±7.55	97.79±8.78	87.99±7.19	95.90±3.43	94.34±7.60	96.61±5.18	94.41±6.30	97.17±7.23	85.97±3.98	97.83±3.65	**98.95±1.64**
12	78.21±5.65	84.54±7.94	87.81±6.87	67.84±3.24	67.08±12.68	84.99±11.54	92.81±8.17	98.09±3.26	73.40±20.85	73.46±20.03	74.38±12.81	96.82±4.14	**99.12±1.08**
13	96.48±1.64	99.65±0.92	96.24±2.72	83.99±1.68	82.24±8.24	94.06±9.57	99.66±0.51	99.01±.59	92.65±5.28	82.52±31.69	97.55±7.36	99.58±0.20	**99.93±0.20**
OA	77.78±1.59	79.65±1.50	82.14±4.03	74.30±18.54	70.06±4.89	88.45±4.29	87.31±5.36	90.60±2.99	75.01±5.86	84.38±12.68	77.10±1.53	93.44±1.86	**96.25±2.58**
AA	71.76±1.60	73.44±1.74	76.85±2.75	70.55±16.81	64.68±4.29	85.06±2.83	84.98±6.45	87.20±4.79	72.18±2.67	84.85±8.69	73.15±1.87	90.76±2.25	**93.79±4.44**
Kappa	75.25±1.75	77.34±1.66	80.07±3.74	71.38±20.56	66.50±5.53	87.10±4.83	85.86±5.96	89.52±3.33	72.19±6.40	82.70±13.94	74.48 ±1.73	92.68±2.07	**95.82±2.87**

TABLE VIII

QUANTITATIVE CLASSIFICATION RESULTS OF THE HONGHU DATASET WITH TEN LABELED TRAINING SAMPLES PER CLASS. BEST RESULTS ARE SHOWN IN BOLD

Class	RF	SVM	ContextNet	SSAN	RSSAN	SSAtt	A2S2K	SSTN	SSSAN	CVSSN	RSEN	ETF	Proposed
1	83.14±7.85	93.86±3.85	87.44±5.97	95.67±2.76	87.62±7.30	96.17±2.36	86.80±18.61	95.96±4.62	93.02±6.96	96.09±3.51	88.07±5.78	**98.88±1.11**	98.71±1.45
2	45.08±7.51	50.34±7.32	48.22±9.76	62.55±9.20	49.05±8.53	69.12±6.41	54.93±8.08	66.91±12.67	61.12±8.51	**71.27±10.78**	70.14±5.84	70.14±11.79	69.15±6.67
3	82.04±4.79	85.33±6.87	85.46±5.24	89.29±5.4	79.18±8.99	93.95±2.10	84.85±12.95	94.24±6.27	84.84±10.29	93.19±3.02	78.10±3.70	**96.01±4.31**	95.50±3.53
4	96.63±0.57	96.62±0.60	98.04±1.00	99.23±0.55	97.51±1.34	99.24±0.56	98.27±1.14	99.36±0.40	98.68±0.59	98.93±0.64	94.47±2.89	99.36±0.41	**99.52±0.22**
5	11.82±.97	13.88±2.75	11.69±5.62	20.33±8.34	11.83±5.37	18.50±7.33	20.74±6.67	43.86±24.22	20.11±13.92	31.87±17.23	**76.64±11.00**	45.04±16.81	46.57±19.64
6	80.93±4.19	86.73±3.91	80.87±3.49	93.14±2.82	85.72±4.00	92.68±2.19	89.28±7.20	96.70±1.95	91.47±3.43	94.18±2.64	85.72±4.79	95.92±2.16	**97.98±1.00**
7	59.45±14.50	68.51±9.62	60.95±11.15	75.63±11.95	65.33±13.79	84.66±6.75	71.63±16.06	**90.75±3.72**	74.84±12.35	86.78±5.42	49.18±12.78	90.45±4.04	**90.75±4.93**
8	6.22±1.56	7.79±2.67	12.31±2.78	16.75±3.04	11.74±2.23	22.19±4.36	27.03±11.42	32.26±8.22	13.32±2.84	32.26±8.34	37.41±3.94	**40.98±12.76**	35.86±6.59
9	80.96±7.77	92.89±4.84	86.66±6.72	95.41±4.15	85.39±7.76	97.40±2.18	81.77±22.54	98.89±0.81	93.71±6.00	96.68±1.71	92.69±2.59	98.10±1.01	**99.04±0.68**
10	27.78±10.00	37.33±11.42	46.86±7.13	58.62±8.07	49.62±9.32	73.43±6.96	73.83±15.31	82.79±10.68	56.96±12.18	82.39±4.96	47.91±8.30	**92.38±4.70**	85.98±10.65
11	17.19±5.49	22.60±8.49	31.02±6.21	36.98±10.40	27.88±6.73	53.94±11.07	31.34±7.36	64.05±11.78	40.33±10.60	62.82±14.19	41.09±10.09	**76.50±11.58**	72.88±8.62
12	36.12±8.32	36.01±8.85	42.60±6.01	57.40±7.51	41.56±5.96	60.78±8.24	51.63±14.81	67.23±10.39	37.82±11.83	64.76±10.77	58.87±10.21	**71.67±13.45**	69.03±10.39
13	51.96±6.94	54.06±6.90	67.85±11.05	71.99±8.95	65.48±7.91	75.81±5.69	73.25±12.07	73.41±8.15	60.89±7.15	78.60±7.12	59.31±6.72	**84.69±7.31**	78.50±9.76
14	44.71±19.29	57.14±18.12	36.08±9.51	65.85±10.96	41.78±12.53	73.65±7.22	68.41±12.08	85.13±10.65	65.72±11.71	76.37±8.62	68.82±8.42	**91.12±6.35**	89.96±4.32
15	3.00±1.21	6.48±4.30	6.57±2.66	54.40±26.75	9.88±5.10	47.82±12.90	8.09±3.51	67.49±22.44	29.45±18.85	67.12±11.55	**90.71±11.55**	78.29±19.55	77.74±11.82
16	75.82±7.02	83.65±3.53	73.67±7.60	88.04±6.28	71.69±7.95	89.59±3.64	86.98±11.63	97.28±4.05	88.98±5.05	95.09±1.66	61.59±12.59	96.66±3.13	**99.04±0.82**
17	37.67±3.13	43.52±5.99	32.22±6.53	48.36±12.94	34.81±4.37	50.32±8.82	59.36±13.83	70.28±11.48	53.02±16.05	76.81±11.64	85.00±10.60	**88.03±9.27**	84.97±9.19
18	13.90±2.51	18.26±3.45	26.46±6.32	36.01±11.37	20.59±7.66	43.61±7.75	52.05±18.05	58.51±16.08	27.79±9.77	45.27±13.77	**79.59±6.21**	64.97±10.98	69.25±11.43
19	42.67±9.31	59.98±8.04	68.82±9.92	75.27±9.60	51.49±16.38	79.36±7.15	76.17±9.23	85.63±5.44	74.67±11.43	80.37±8.03	66.24±5.12	85.25±5.95	**90.11±3.80**
20	11.40±3.51	25.82±6.27	47.43±13.26	69.87±17.19	42.07±9.89	64.56±8.45	39.36±17.35	77.24±5.27	55.28±11.96	65.60±11.62	**85.69±5.95**	77.83±4.79	85.25±5.28
21	8.87±10.70	8.31±1.15	8.25±2.37	13.04±4.41	7.38±2.13	12.79±3.04	19.11±8.57	22.33±7.66	10.64±3.02	20.09±6.95	**83.66±9.20**	30.05±8.92	33.67±7.57
22	19.71±3.91	19.64±4.61	20.08±7.11	33.13±9.39	22.51±8.83	33.90±9.32	23.56±8.61	62.97±17.33	34.22±10.21	43.89±17.10	**68.51±20.03**	53.93±16.53	67.92±12.43
OA	51.46±3.89	59.80±3.29	62.28±6.32	73.33±7.77	61.17±8.42	76.99±4.47	67.35±8.08	85.05±3.78	69.08±8.62	81.92±4.01	80.13±1.74	88.28±2.74	**88.86±2.11**
AA	42.59±1.79	48.58±1.24	49.30±1.94	61.68±3.50	48.19±3.64	65.16±1.81	58.11±4.34	74.24±1.98	57.58±4.81	70.92±2.65	71.35±1.57	78.47±3.07	**78.97±1.46**
Kappa	44.61±3.42	52.99±3.05	55.77±6.12	68.17±7.77	54.58±8.14	72.26±4.92	61.66±8.58	81.57±4.29	63.33±8.72	77.86±4.56	75.13±2.04	85.49±3.24	**86.16±2.51**

Besides our algorithm, the ETF algorithm achieves the best performance, leveraging preallocated classifiers to enhance hyperspectral classification. It is worth noting that the RSEN algorithm, also designed for small samples, achieves high classification performance for certain classes across all four datasets, especially when other algorithms exhibit lower accuracy. However, the overall OA and Kappa values of this algorithm are not particularly ideal.

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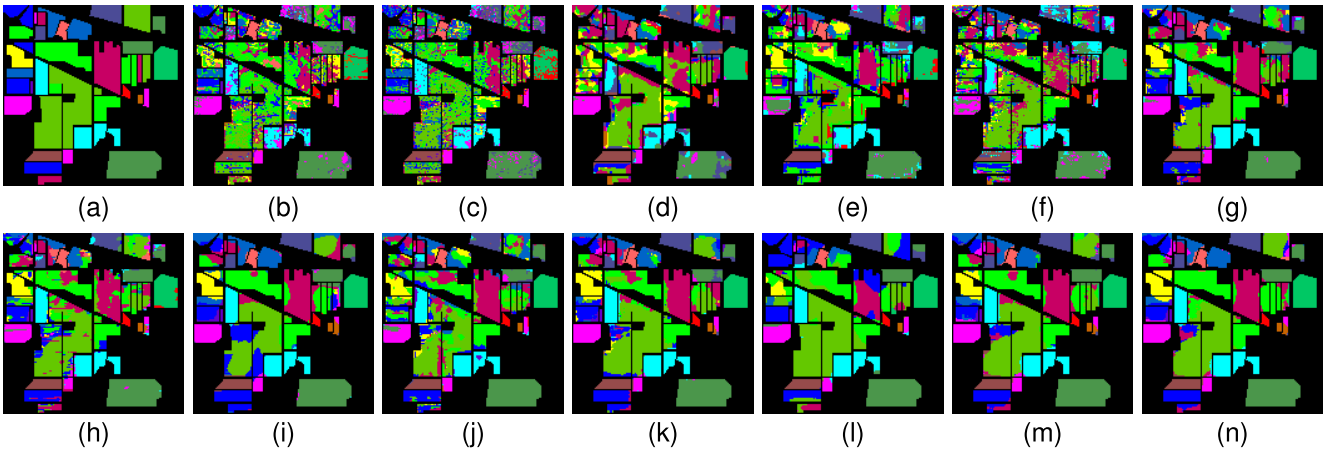


Fig. 5. Classification maps for the IPs dataset with ten labeled training samples per class. (a) Ground-truth map. (b) RF. (c) SVM. (d) ContextNet. (e) SSAN. (f) RSSAN. (g) SSAtt. (h) A2S2K. (i) SSTN. (j) SSSAN. (k) CVSSN. (l) RSEN. (m) ETF. (n) Proposed.

Comparing the results of these four quantitative tables with the earlier tables on test and training sample numbers, we find that for imbalanced samples, especially for some classes with fewer samples, RSEN can achieve higher accuracy. This is mainly because RSEN learns a fixed number of pseudolabeled samples for each class during pseudolabel learning, which may utilize many unlabeled samples during training, resulting in higher accuracy for these classes. However, for classes with a larger number of samples, the relatively fewer training samples may not fully exploit the accuracy of these classes, leading to inadequate classification performance.

Additionally, in addition to our proposed algorithm and the ETF algorithm, the SSTN and CVSSN algorithms demonstrate good results on three datasets, except for the KSC dataset. On the KSC dataset, SST and SSAtt achieve the best classification performance alongside our algorithm. Overall, our algorithm consistently achieves the highest classification performance across all four datasets.

In addition, it is worth noting that the IP and Honghu datasets exhibit more severe class imbalance issues compared to the UP and KSC datasets, making classification tasks more challenging for these two datasets. In these two datasets, due to class imbalance problems, the AA of almost all algorithms is much lower than the OA. However, in the IP dataset, the AA value of the RSEN algorithm is relatively high, mainly because some classes with few samples achieve extremely high classification accuracy. Overall, the proposed algorithm consistently achieves the best classification performance across all four datasets.

We visualized the classification results. Due to the limited number of training samples used in this study, the classification accuracy of most algorithms is relatively low. Therefore, we only show the visualization of the classification results with labeled samples. The classification results without labeled samples cannot determine whether the classification is correct, so we set them to a black background to better identify the visual differences in classification results between different algorithms. The classification results on the four datasets are shown in Figs. 5–8.

Specifically, on challenging datasets like the IPs dataset, RF and SVM algorithms utilized only spectral information without considering any spatial information. Additionally, the hyperspectral images themselves may exhibit isotropic and anisotropic pixel aliasing phenomena, further increasing the difficulty of obtaining accurate classification results. Therefore, these algorithms perform relatively poorly in terms of classification performance. On the other hand, we observed some classification noise in the edge regions of the classification maps generated by ContextNet, SSAN, RSSAN, SSAtt, and SSSAN. This noise may be due to the limited feature expression capability of these algorithms when dealing with a small number of samples. In contrast, algorithms designed for small samples, such as RSEN, ETF, SSTN, and CVSSN, exhibit smoother and clearer classification results on the maps. Compared to the results of these algorithms, our proposed algorithm shows better classification map results and is closer to the ground truth.

Similarly, when classifying the Pavia dataset, a significant number of misclassification issues were observed with two machine learning algorithms. On this dataset, besides our proposed algorithm, ETF algorithm demonstrated the best visual performance. In comparison, SSAtt and CVSSN achieved satisfactory classification results. Notably, in the bottom right corner of the classification maps, our algorithm exhibited less noticeable noise compared to these algorithms, indicating higher classification accuracy. At the same time, RSEN showed a significant amount of misclassification in this area, leading to an overall decrease in accuracy. This could be attributed to the poor quality of pseudolabels obtained during pseudolabel learning due to the variations in spectral signatures of the same object in hyperspectral images, resulting in reduced classification accuracy. Among all algorithms, our proposed algorithm exhibited the fewest classification errors and the highest classification accuracy.

It is worth noting that the Honghu dataset comprises 22 feature classes, some of which share certain similarities in their spectral characteristics. Moreover, hyperspectral images commonly exhibit phenomena where the same type of land

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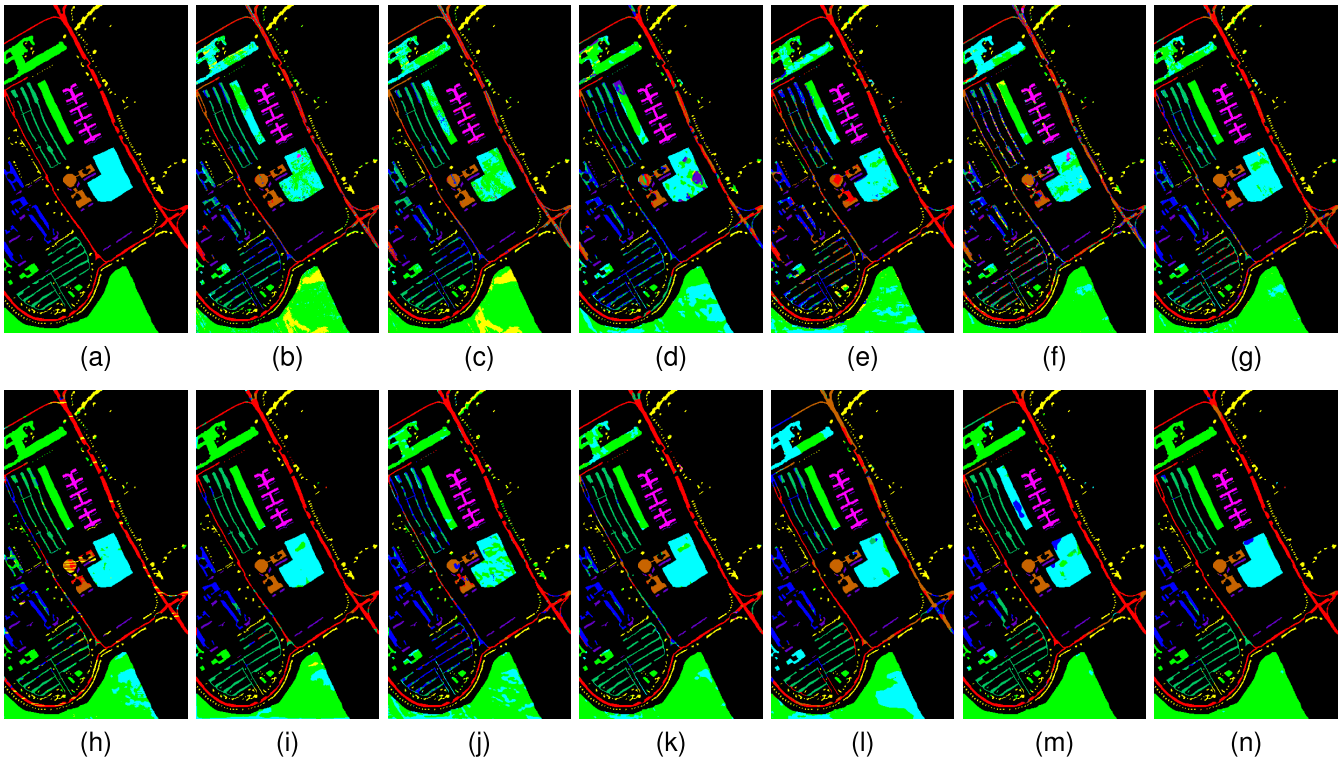


Fig. 6. Classification maps for the PU dataset with ten labeled training samples per class. (a) Ground-truth map. (b) RF. (c) SVM. (d) ContextNet. (e) SSAN. (f) RSSAN. (g) SSAtt. (h) A2S2K. (i) SSTN. (j) SSSAN. (k) CVSSN. (l) RSEN. (m) ETF. (n) Proposed.

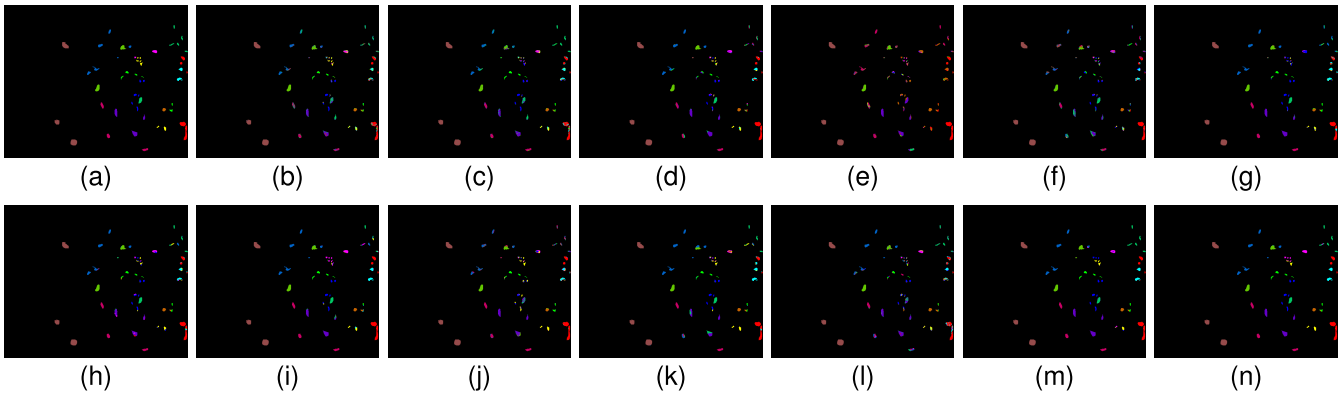


Fig. 7. Classification maps for the KSC dataset with ten labeled training samples per class. (a) Ground-truth map. (b) RF. (c) SVM. (d) ContextNet. (e) SSAN. (f) RSSAN. (g) SSAtt. (h) A2S2K. (i) SSTN. (j) SSSAN. (k) CVSSN. (l) RSEN. (m) ETF. (n) Proposed.

cover may have different spectral signatures and different types of land cover may share similar spectral characteristics. This phenomenon poses challenges to classification algorithms, such as RF and SVM, which rely solely on spectral features, resulting in numerous misclassifications throughout the classification area. In comparison to other benchmark algorithms, our method demonstrates significantly fewer classification errors, highlighting its advantages, especially under limited sample conditions. Particularly noteworthy is the smoother representation with fewer misclassifications observed in the middle yellow area, which is more evident in the classification results of the ETF algorithm. Apart from this algorithm, the overall classification map of our proposed method appears smoother compared to others. On this dataset, our proposed algorithm achieves classification performance comparable to that of the ETF algorithm, resulting in visually similar classification maps for most categories.

2) Efficiency Analysis: We conducted efficiency comparisons among different methods and summarized the results in Table IX, which includes metrics such as training time, running time, number of model parameters, and computational cost. Regarding runtime, traditional methods such as RF and SVM typically require less time during both model training and testing phases. Additionally, the training process is generally faster than testing, influenced by sample size. Conversely, deep learning-based algorithms exhibit the opposite trend, with longer training times but shorter testing times.

In terms of training time, our algorithm does not show a significant advantage among deep learning algorithms. However, it performs faster during the testing phase compared to other

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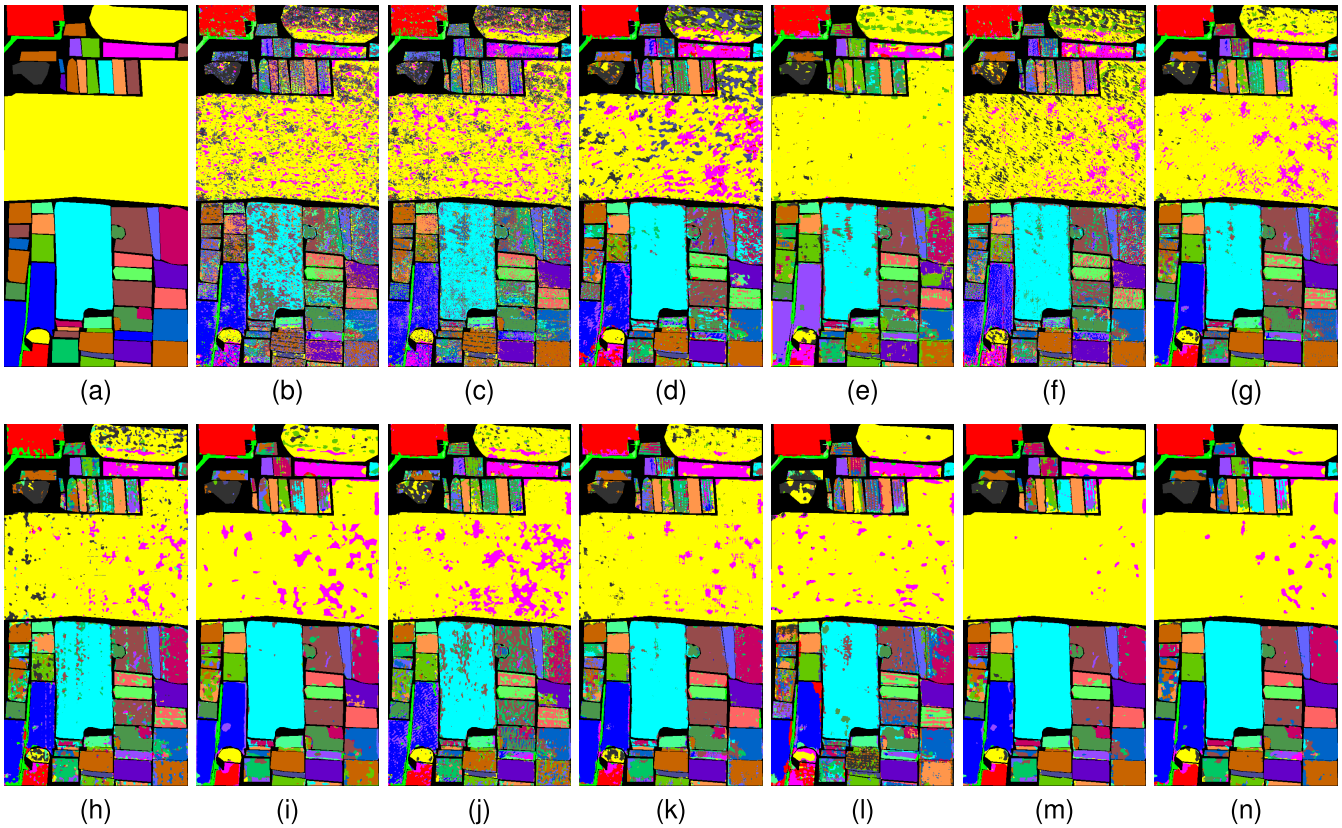


Fig. 8. Classification maps for the Honghu dataset with ten labeled training samples per class. (a) Ground-truth map. (b) RF. (c) SVM. (d) ContextNet. (e) SSAN. (f) RSSAN. (g) SSAtt. (h) A2S2K. (i) SSTN. (j) SSSAN. (k) CVSSN. (l) RSEN. (m) ETF. (n) Proposed.

TABLE IX

COMPARISONS ON MODEL EFFICIENCY OF DIFFERENT METHODS ON IP, UP, KSC, AND HONGHU DATASETS

Dataset	metrics	RF	SVM	ContextNet	SSAN	RSSAN	SSAtt	A2S2K	SSTN	SSSAN	CVSSN	RSEN	ETF	Proposed
IP	T_{train}	1.27±0.21	**0.04±0.03**	177.35±4.46	503.56±26.89	147.27±4.97	138.06±3.75	567.34±19.49	167.84±11.768	552.60±32.65	125.29±5.63	647.17±2.56	15.36±0.33	42.74±1.72
	T_{test}	**1.09±0.17**	1.58±0.24	7.24±1.15	11.97±1.52	10.88±0.77	10.53±1.09	4.54±0.10	11.58±1.64	106.85±4.11	6.30±0.71	1.03±0.07	1.09±0.01	1.21±0.01
	Params(M)	-	-	1.21	87.68	0.17	0.60	0.37	**0.02**	0.31	0.26	1.28	0.09	0.07
	FLOPs(M)	-	-	127.8	10548.91	11.76	22.50	256.71	**2.67**	26.74	31.46	12.53	10.66	8.15
UP	T_{train}	0.76±0.18	**0.01±0.01**	160.37±5.19	182.47±19.31	116.06±14.32	103.61±6.24	287.58±12.01	162.51±24.57	212.98±7.66	93.18±3.92	644.56± 7.92	4.76±0.10	38.71±1.79
	T_{test}	2.23±0.33	1.32±0.22	48.22±9.76	8.37±1.22	1.70±0.06	13.53±1.18	5.12±0.34	6.12±0.44	64.17±3.65	12.05±2.11	6.68±1.08	**0.90±0.00**	1.27±0.03
	Params(M)	-	-	0.70	25.68	0.11	0.53	0.22	**0.02**	0.28	0.25	1.17	0.09	0.07
	FLOPs(M)	-	-	74.52	2960.99	8.32	15.75	131.68	**2.09**	17.51	29.90	12.44	9.47	7.05
KSC	T_{train}	0.82±0.06	**0.01±0.01**	132.18±8.46	334.22±21.55	137.19±4.77	138.06±2.65	470.86±7.19	161.34±6.77	78.92±13.22	91.66±11.63	245.63±5.03	9.17±0.68	39.66±1.08
	T_{test}	**0.38±0.13**	0.43±0.10	2.35±0.06	4.73±0.34	14.09±1.93	10.07±0.74	14.55±1.28	5.86±0.28	419.45±13.23	5.75±0.66	6.33±1.37	1.07±0.02	1.20±0.01
	Params(M)	-	-	1.07	68.75	0.15	0.58	0.33	**0.02**	0.30	0.26	1.25	0.09	0.06
	FLOPs(M)	-	-	114.46	8237.33	10.90	20.84	225.45	**2.52**	24.46	31.08	12.51	10.47	7.96
HH	T_{train}	1.84±0.30	**0.07±0.05**	79.68±3.345	783.70±16.90	61.84±2.63	79.02±6.21	843.37±19.75	94.04±7.56	271.18±26.20	68.90±9.76	386.66±3.72	9.70±0.33	61.28±.33
	T_{test}	47.94±3.46	50.34±7.32	1.93±0.05	2.90±0.05	3.29±0.05	4.72±0.15	5.89±0.19	7.26±0.53	32.15±1.06	5.05±0.43	7.96±0.39	1.44±0.33	**1.54±0.03**
	Params(M)	-	-	1.59	156.36	0.22	0.64	0.48	**0.02**	0.34	0.27	1.26	0.09	0.07
	FLOPs(M)	-	-	166.26	18921.61	14.24	27.40	347.89	**3.08**	33.35	32.60	13.61	10.68	8.28

deep learning-based methods. This can be attributed to the inclusion of randomly generated samples during the training process in our algorithm, which extends the training time to some extent.

We also compared the number of model parameters among different algorithms. From the table, it is evident that our proposed algorithm has a lower number of parameters compared to others. The parameter count of our algorithm is only slightly higher than RSSAN and SSTN algorithms. Additionally, our algorithm demonstrates a smaller model size compared to the state-of-the-art algorithm CVSSN. Moreover, in terms of computational cost, our algorithm shows relatively lower floating-point operations. Among the compared deep learning algorithms, our algorithm ranks third in terms of computational requirements. Overall, considering these results along with the previous table, we conclude that our algorithm achieves superior classification performance while maintaining a relatively small model size and fewer parameters.

It is worth noting that the ETF algorithm, despite having a larger model and parameter count, exhibits training time significantly better than the proposed algorithm. This could be attributed to the classifier being preassigned in this algorithm, thus not requiring backpropagation to update classifier parameters, resulting in improved classification accuracy. On the other hand, the RSEN algorithm, designed for small sample

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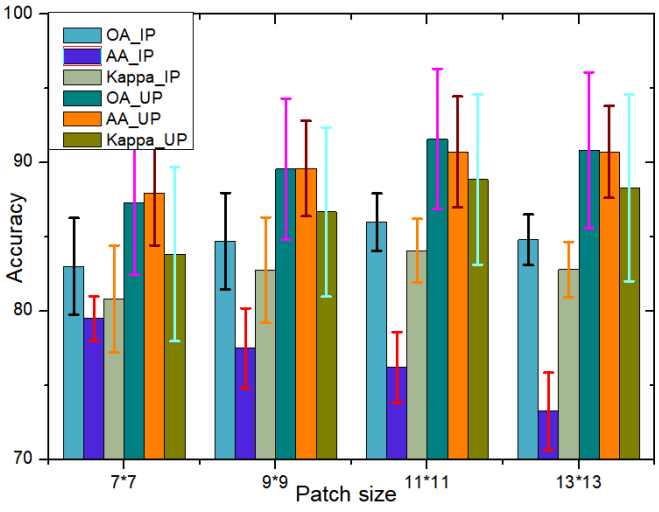


Fig. 9. Accuracy of different patch size.

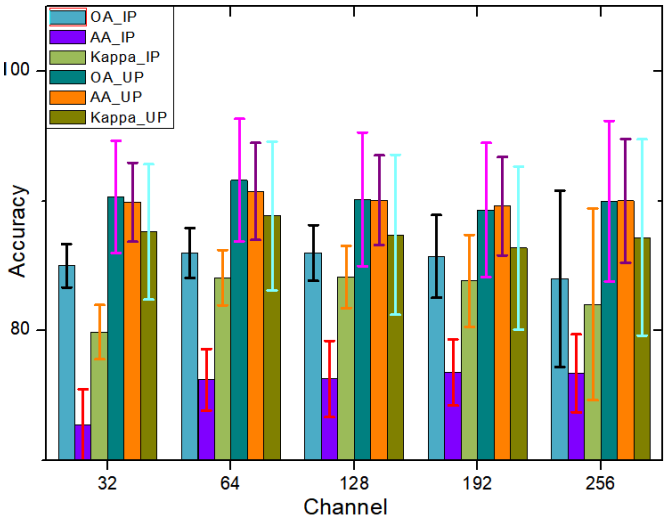


Fig. 10. Accuracy of different channel numbers.

classification, although having relatively lower testing time, requires significant training time. This is mainly because the algorithm needs to utilize many unlabeled samples for pseudolabel learning, which consumes more time compared to other algorithms.

E. Partial Parametric Analysis

In the experiment, we also investigated the impact of different parameter settings on the final results, particularly the size of the samples and the channels of the feature. We conducted experiments on two datasets, IP and UP, to analyze the classification accuracy achieved under different conditions.

In Fig. 9, we present the classification results of different patch sizes on two datasets. The experiments show that when the patch size is 11*11, the best performance is achieved on both datasets. When analyzing the impact of the number of feature channels on our experimental results, we kept all other settings constant and varied the middle feature channel (initially set to 64) to different values: 32, 64, 128, 192, and 256, respectively. The results are illustrated in Fig. 10. It is noteworthy that when the number of channels is 32, the classification accuracy is relatively low across multiple indicators, while at 64 channels, we achieved relatively favorable results on several indicators. However, with a further increase in the number of channels, the OA, AA, and Kappa coefficients did not continue to improve; instead, they decreased. This phenomenon can be attributed to the increasing complexity of the network, leading to a larger number of parameters that need to be learned. Due to the limited training samples, larger networks become more challenging to train and are more susceptible to problems like overfitting.

F. Ablation Study

To validate the effectiveness of all our modules, we conducted ablation experiments on four datasets while keeping the experimental parameter settings consistent. The results of the ablation experiments are presented in Table X. In the table, "Sample" refers to the online sample augmentation training module, "Att-Res" refers to the attention-enhanced residual module we proposed, and "SC-Att" refers to the spatial-channel attention module we introduced.

TABLE X
COMPARISONS ON MODEL EFFICIENCY OF DIFFERENT METHODS ON IP, UP, KSC, AND HONGHU DATASETS

Dataset	Sample	Att-Res	SC-Att	OA	AA	Kappa
IP	✓	✓	✓	**85.98±1.93**	76.20±2.38	**84.08±2.13**
	✓	✓	-	85.11±1.15	**76.78±2.30**	83.12±1.32
	✓	-	✓	83.70±1.67	74.37±2.73	81.56±1.91
	-	✓	✓	85.10±1.61	75.04±2.69	83.14±1.78
	-	-	✓	84.10±3.19	75.76±2.34	82.01±3.59
	-	✓	-	84.30±1.52	75.43±2.34	83.03±1.66
UP	✓	✓	✓	**91.29±4.74**	**90.73±3.74**	**88.86±5.74**
	✓	✓	-	90.47±5.56	90.47±3.62	87.81±6.79
	✓	-	✓	90.50±3.62	90.48±3.10	87.90±4.41
	-	✓	✓	89.75±6.62	90.51±4.19	87.02±7.91
	-	-	✓	90.53±4.40	89.77±3.62	89.23±5.55
	-	✓	-	90.28±3.71	87.46±3.72	87.69±5.58
KSC	✓	✓	✓	**96.25±2.58**	93.79±4.44	**95.82±2.87**
	✓	✓	-	95.80±2.61	**94.32±2.03**	95.31±2.91
	✓	-	✓	95.80±1.51	93.25±2.23	95.32±1.68
	-	✓	✓	95.55±2.21	92.88±3.00	95.04±2.46
	-	-	✓	95.23±2.53	92.79±2.65	94.99±2.55
	-	✓	-	95.38±2.91	92.76±2.15	95.00±1.88
HH	✓	✓	✓	**88.86±2.11**	**78.97±1.46**	**86.16±2.51**
	✓	✓	-	87.40±2.88	78.40±2.07	84.43±3.39
	✓	-	✓	87.05±2.75	77.35±1.53	84.06±3.20
	-	✓	✓	88.20±2.34	78.39±2.61	85.33±2.79
	-		✓	87.59±2.38	78.50±2.35	84.54±3.15
	-	✓	-	87.08±1.95	78.36±2.46	84.15±2.12

In the ablation experiment of the attention-enhanced residual module, we excluded the attention module within this module, while keeping the other parts unchanged. For the ablation experiments involving the two outer modules, the respective modules were removed entirely. In the table, a checkmark ✓ indicates the inclusion of its corresponding module. Please note that all ablation experiments are the average results of ten randomized experiments. The table presents the performance metrics for each configuration, allowing us to

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analyze the impact of each module on the overall classification performance.

From the table, it is evident that the three indicators proposed by us have led to significant improvements in classification accuracy across all four datasets. Let us take the UP dataset as an example: the OA of the three modules improved to 0.77, 0.45, and 0.45, respectively. This demonstrates the effectiveness of our proposed modules in enhancing classification performance. However, it is important to note that in the evaluation of the IP dataset, our model showed a slight decrease in AA. This discrepancy may be attributed to the presence of noise in this particular dataset, which can affect the classification results to some extent. Despite this minor drop in AA, our model still achieves competitive results on the IP dataset, reaffirming the overall effectiveness of our proposed approach. Overall, the ablation experiments provide strong evidence that the integration of our proposed modules significantly enhances the classification accuracy across different datasets, reaffirming the utility of our approach in hyperspectral image classification tasks.

IV. Conclusion and Discussion

In this article, we aimed to address the issue of insufficient sample size in hyperspectral classification by introducing a novel online sample augmentation technique. Building upon this technique, we proposed a hyperspectral classification network called ReSC-net, which integrates a residual module for attention enhancement and a spatial-channel attention module. The experiments validate the effectiveness of our sample augmentation technique and network design.

While our method demonstrates significant advantages in addressing the issue of limited samples in hyperspectral classification, it also presents some limitations and challenges. First, although our online sample augmentation technique proves highly effective in enhancing sample diversity and improving network performance, it may incur higher computational costs and time consumption during training. Additionally, our network architecture may not be sufficiently adaptable to certain types of datasets, resulting in decreased performance or inadequate generalization ability. Moreover, the performance of our method may be constrained when dealing with extremely limited samples, particularly when the number of labeled samples is very small, hindering our method from fully realizing its potential.

In comparison to the RSEN method, our approach benefits from introducing an innovative online sample augmentation technique and designing a hyperspectral classification network that integrates residual modules and spatial-channel attention modules. These characteristics grant our method significant performance advantages in addressing the issue of limited samples in hyperspectral classification. However, the RSEN algorithm utilizes a large number of unlabeled samples during the construction of the training set, which are readily available in practical scenarios. Our proposed algorithm fails to fully leverage the abundant unlabeled samples present in hyperspectral images. Leveraging these unlabeled samples, combined with semi-supervised and weakly supervised algorithmic theories, to further enhance the classification performance of hyperspectral images under limited sample conditions, is a key focus of our future research. Additionally, algorithms such as CVSSN demonstrate the utilization of spatial and spectral information, whereas this article does not extract information from individual pixel spectra, which may partially hinder the performance of the proposed algorithm. Therefore, another research direction for us is to better integrate spectral information into the proposed algorithmic framework.

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